Michael Sprague

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TECHNICAL SUMMARY

Detail-oriented IT and systems specialist with specialized experience in custom PC architecture, hardware diagnostics, and relational data management. Currently expanding into AI engineering and large language model code evaluation, blending practical hardware-level troubleshooting with formal training from Columbia University. Proven track record of high academic performance and technical analysis, with a focus on system reliability, performance optimization, and clear technical communication.

TECHNICAL SKILLS

- **Hardware & Troubleshooting:** Custom PC assembly, GPU/CPU installation, component-level diagnostics, thermal management, peripheral configuration, and hardware lifecycle documentation.
- **Systems & Tools:** Windows 10/11 Pro administration, Microsoft 365, Microsoft Access/VBA for asset tracking, Linux CLI (foundational), Python, hardware diagnostic tools, and AI upscaling/model evaluation workflows.
- **Operations & Networking:** IT inventory management, documentation of help guides/asset lists, Ethernet/IP concepts, and user access/permission support.

PROFESSIONAL EXPERIENCE

AI Technical Specialist (Contract) | *DataAnnotation.tech* | **Jan 2026 – Present**

- Applies root-cause investigation skills to identify and document technical inaccuracies in complex AI-generated code, focusing on Python and SQL data integrity.
- Conducts high-level Reinforcement Learning from Human Feedback (RLHF) to refine large language models (LLMs).
- Ranked in the top 10% of technical contractors based on quality of technical analysis, accuracy of code evaluation, and adherence to rigorous system requirements.

Technical Intern | *Holy Cross College / DePaul Academy* | **2023**

- Engineered custom Microsoft Access databases and VBA automation to manage hardware inventory, asset tracking, and service records.

- Improved hardware tracking data integrity, significantly reducing administrative latency for equipment maintenance and device offboarding.

Writing Center & Mathematics Tutor | *Holy Cross College* | **2021 – 2025**

- Translated complex technical and mathematical concepts into clear, actionable instruction for users with varying levels of technical literacy.

Clerical Assistant | *Oakland City University* | **2017 – 2020**

- Maintained and updated Microsoft Access databases supporting university administrative operations and records management.
- Developed automated templates and workflow tools that improved documentation consistency, reducing manual processing time for staff requests.

Hardware Technician & System Builder | *Independent / Personal Projects*

- Designed, assembled, and maintained 20+ high-end custom desktop computer systems, ensuring optimal thermal performance and long-term structural integrity.
- Conducted component validation, hardware stress testing, and OS deployment to swiftly isolate and resolve hardware defects, seating issues, or software conflicts.
- Managed localized system updates, backups, user account configurations, and applied cybersecurity best practices to secure client environments.

AI Video Upscaling Study (Vivi's Dream) | *Independent AI Project* | **2026**

- Built and published a side-by-side technical comparison showcase evaluating the practical limitations and visual fidelity of modern AI upscaling pipelines on legacy video.
- Tested multi-stage model workflows to isolate failure modes, artifacting, and loss of detail when processing low-resolution source media.
- Documented performance and image clarity trade-offs, demonstrating where current automated upscaling tools break down under constrained input data.

EDUCATION & CERTIFICATIONS

Columbia University | *AI Engineering Certificate*

- Pathways to Applied AI Solutions Engineering | Feb 2026 – Present

Holy Cross College | *Bachelor of Arts in Business*

- GPA: 3.98 | Summa Cum Laude | Delta Epsilon Sigma

- *Capstone Project:* The Economics of Quantum Computing – Analyzed infrastructure requirements and scalability of high-performance computing architectures.

Certifications

- **CompTIA A+** | *In Progress*
- **Google IT Support Professional Certificate** | *2026*
- **NVIDIA AI Infrastructure and Operations Fundamentals** | *2026*